Department of Computer Science & Cybersecurity

Digital Forensics — Course Study Notes

Unit 2: Digital Evidence & Acquisition

Topic Focus: 2.4 Evidence Handling Procedures & 2.5 Data Acquisition Techniques (Live & Dead)

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1 Introduction

In digital forensics, the integrity of digital evidence is paramount. Any alteration, deletion, or contamination of data during the forensic lifecycle can render it inadmissible in a court of law. This unit focuses on two critical operational domains of forensic investigations:

1. **Evidence Handling Procedures (2.4):** The administrative and operational protocols followed from the moment evidence is identified until it is presented in court.
2. **Data Acquisition Techniques (2.5):** The technical methodologies utilized to capture and preserve digital assets from dynamic (Live) or static (Dead) systems.

2 2.4 Evidence Handling Procedures

The core objective of evidence handling is maintaining integrity, preventing contamination, and ensuring a verifiable **Chain of Custody**.

2.1 Identification and Collection

Before touching any system, an investigator must identify all potential sources of evidence and document the scene thoroughly.

- **Initial Assessment:** Determine the scope of the incident. Identify computers, mobile devices, servers, cloud storage, external media, and IoT devices.
- **Scene Documentation:** Take high-resolution photographs of the physical scene, system screens, cabling, and serial numbers. Document the state of the machine (On, Off, Suspended, or Locked).
- **Collection Order:** Collect physical items in a way that minimizes contamination. Use clean latex gloves to prevent transferring physical fingerprints or DNA to devices.

2.2 The Chain of Custody

The **Chain of Custody (CoC)** is a chronological, written record of custody, control, transfer, analysis, and disposition of physical or electronic evidence. If the defense can prove a gap in the CoC, the evidence is highly likely to be thrown out of court.

Every entry in the Chain of Custody must document:

1. **Who** took possession of the evidence.

2. **What** the evidence item is (description, model, serial number).
3. **When** the transfer occurred (precise date and time).
4. **Where** the evidence was secured or transported.
5. **Why** the transfer occurred (e.g., "Transport to lab for imaging").

Standard Chain of Custody Template Form

Case Number: 2026-DF-0941 **Investigator:** Det. Alex Chen
Item No: 01 **Description:** Samsung 1TB External SSD (S/N: MZ-77E1T0B)

Date/Time	Released By	Received By	Purpose / Action Taken	Location
16/07/2026 10:15	Scene Recovery	Det. Alex Chen	Collected at crime scene	Server Rm, Bldg 4
16/07/2026 11:45	Det. Alex Chen	Evid. Custodian	Logged into secure storage	Forensic Lab Vault
17/07/2026 09:00	Evid. Custodian	Analyst S. Kumar	Bit-stream imaging (E01)	Lab Workstation 3

2.3 Packaging, Transportation, and Storage

Physical and environmental hazards can easily corrupt digital components. Proper packaging and transportation are crucial.

- **Anti-Static Packaging:** Hard drives and circuit boards are highly sensitive to Electrostatic Discharge (ESD). They must be packed in anti-static Faraday bags.
- **Faraday Shielding:** For mobile phones, tablets, or wireless-enabled devices, packaging in a **Faraday Bag** is mandatory. This blocks all incoming/outgoing radio waves (cellular, Wi-Fi, Bluetooth), preventing remote-wipe commands or incoming data modification.
- **Transportation Safety:** Secure devices in padded, shock-absorbent containers. Avoid extreme temperature fluctuations during transport, as condensation can damage internal electronics.
- **Secure Storage:** Evidence must be stored in a climate-controlled, secure room or vault with restricted access. Every entry and exit to the storage facility must be logged.

3 2.5 Data Acquisition Techniques

Data acquisition is the process of retrieving digital information from physical storage media or volatile memory in a forensically sound manner. Acquisition is broadly categorized into **Live (Volatile)** and **Dead (Non-volatile)** acquisition.

3.1 Order of Volatility

When acquiring evidence, investigators must prioritize the collection of data based on how quickly it might be lost. This is defined by **RFC 3227 (Order of Volatility)**:

1. **Registers, Cache, and Active CPU States** (Changes in nanoseconds)
2. **Routing Tables, ARP Cache, Process Table, Kernel Statistics, RAM** (Changes in microseconds)
3. **Temporary File Systems / Swap Space** (Changes in minutes)
4. **Hard Disk Drives (HDD/SSD) and Storage Media** (Persists until overwritten)
5. **Remote Logging and Monitoring Data** (Persists on secondary servers)
6. **Physical Configuration and Network Topology** (Relatively static)

3.2 Live Acquisition (Volatile Data)

Live acquisition captures data from a system that is actively **powered on** and running.

3.2.1 Characteristics & Use Cases

- Captured when encrypted drives (BitLocker, VeraCrypt) are unlocked in memory.
- Essential for retrieving transient states, active network connections, logged-in sessions, and running malware processes.
- Used heavily in Incident Response (IR) where downtime is not acceptable for critical business servers.

3.2.2 Key Challenges

- **The Observer Effect:** Running an acquisition tool on a live system alters the system state (modifies RAM, updates file access times, and runs new processes).
- **Malware Interference:** Rootkits and sophisticated malware can detect forensic tools and actively lie, hide processes, or trigger self-destruct scripts.

3.2.3 Common Live Acquisition Tools

- **FTK Imager Lite:** A portable version of FTK Imager designed to run from an external USB drive to capture RAM and physical disks.
- **DumpIt / Belkasoft RAM Capturer:** Dedicated tools for dumping physical memory (RAM).

- **Volatility Framework:** An advanced command-line tool used to analyze the captured RAM dump offline.

3.3 Dead Acquisition (Static / Post-Mortem)

Dead acquisition takes place when the target machine is **powered off**. It is the most traditional and forensically secure method.

3.3.1 Characteristics & Process

- The target storage drive is physically removed from the machine (where possible).
- The drive is connected to a **Write Blocker** to ensure no write commands reach the evidence media.
- A bit-stream image (exact replica of every sector, including unallocated space, slack space, and deleted files) is created.

3.3.2 Write Blockers

Write blockers are critical to dead acquisition. They allow read commands but block write commands.

- **Hardware Write Blockers:** Physical bridging devices between the forensic workstation and the evidence drive (e.g., Tableau, WiebeTech). Highly reliable.
- **Software Write Blockers:** Operating system-level blocks (e.g., modifying the registry in Windows to block writes to USB devices). Economical but less secure than hardware solutions.

3.3.3 Forensic Image Formats

During a dead acquisition, the copy is stored in specific forensic image formats to guarantee data integrity:

- **Raw (DD/RAW):** A sector-by-sector clone with no metadata or compression. Extremely versatile but takes up a large amount of storage. File extensions are typically `.raw`, `.dd`, or `.img`.
- **Advanced Forensic Format (AFF):** An open-source extensible format that supports metadata, compression, and built-in MD5/SHA-1 hashing.
- **EnCase Evidence File (E01):** The industry standard developed by Guidance Software. It contains the acquisition metadata (case info, examiner name, acquisition date) and embeds cryptographic hashes at regular intervals to detect and report corruption instantly.

3.4 Live vs. Dead Acquisition: Comparative Matrix

Below is a structured comparison to help students understand when to deploy each technique.

Table 1: Comparative Matrix: Live vs. Dead Acquisition

Metric	Live Acquisition (Dynamic)	Dead Acquisition (Static)
System State	Machine is powered on and running	Machine is powered off
Main Focus	RAM, Network Connections, Active Processes	Non-volatile storage (HDD/SSD, USB)
Risk Level	High (Observer effect, malware tampering)	Negligible (Protected by Write Blockers)
Completeness	Captures active/hidden transient states	Captures deleted/unallocated sectors
Admissibility	Subject to challenge; must be meticulously logged	Readily accepted in courts of law
Common Tools	FTK Imager Lite, DumpIt, WinPmem	Hardware Write Blockers, EnCase, FTK

4 2.6 Disk Imaging and Cloning

In digital forensics, replicating data without altering the original media is fundamental. Two primary methods are used to create replica evidence: **Imaging** and **Cloning**. Understanding their differences, mechanics, and validation protocols is essential for forensic examiners.

4.1 Imaging vs. Cloning

While both processes involve copying digital storage media, they serve different operational purposes:

- **Disk Imaging (Forensic Image):** This process creates a single, bit-stream duplicate file (or set of split files, e.g., `.E01` or `.raw`) containing an exact, sector-by-sector copy of the

physical disk. The output is a *file* stored on a formatted target media. It preserves all active files, deleted files, slack space, and unallocated clusters.

- **Disk Cloning:** This process replicates the contents of a suspect source drive sector-by-sector directly onto another physical drive of equal or greater capacity. The resulting target disk becomes a bootable, fully functional replica of the original drive.

4.2 Bit-stream Copy vs. Logical Copy

A critical distinction exists between forensic bit-stream copies and traditional logical or OS-level file copies:

- **Logical Copy:** Only copies active, visible files recognized by the filesystem structure. It ignores unallocated space, system metadata, deleted items, file slack, and partitions unrecognized by the operating system.
- **Bit-stream Copy (Forensic Copy):** Replicates every single bit on the drive sequentially from Sector 0 to the final sector, regardless of the filesystem structure. It captures:
  - **Deleted files:** Data blocks marked as ”free” but not yet overwritten.
  - **File Slack Space:** The unused space between the end of a file and the end of the physical sector/cluster allocating it, which often contains remnants of older data.
  - **Unallocated Space:** Portions of the disk not assigned to any file.
  - **Hidden/System Partitions:** Master Boot Record (MBR), GUID Partition Tables (GPT), Host-Protected Areas (HPA), and Device Configuration Overlays (DCO).

4.3 Forensic Verification & Integrity Validation

To prove in court that an image is a flawless, untampered replica of the suspect media, examiners employ cryptographic hashing algorithms (typically **MD5**, **SHA-1**, or **SHA-256**).

1. **Pre-Acquisition Hash:** A hash value is computed on the original source drive prior to starting the copy process (or during read-pass).
2. **Post-Acquisition Hash:** Once the image file or clone is created, the system calculates a hash of the generated image.
3. **Verification / Math Parity:** If the two hash values match perfectly (e.g., *SHA-256 Source* matches *SHA-256 Target*), it mathematically guarantees that the target image is an exact, 100% match of the source, with zero bits changed.

If even a single bit of the original or copy changes during transport or imaging, a completely different hash value will be generated due to the ”avalanche effect.”

5 2.7 Forensic Tools: FTK Imager and dd

Forensic investigators rely on verified software tools to perform static and live data acquisitions. Two of the most widely used tools in both corporate and law enforcement contexts are **FTK Imager** (GUI-based) and **dd** (command-line-based).

5.1 FTK Imager (AccessData)

FTK Imager is a free, industry-standard tool used for creating forensic images, previewing evidence, and verifying data integrity. It can be run as an installed application or in a portable format (*FTK Imager Lite*) to minimize changes on a live target system.

5.1.1 Core Features & Functions

- **Image Creation:** Supports creating forensic images of physical drives, logical drives, individual folders, or contents of volatile RAM.
- **Forensic Formats:** Can output images in Raw/DD (`.001`), EnCase/Expert Witness Format (`.E01`), Advanced Forensic Format (`.AFF`), and Smart (`.S01`).
- **Image Mounting:** Allows examiners to mount a previously acquired forensic image as a read-only local drive to browse files through Windows Explorer.
- **RAM Capturing:** Volatile memory dump extraction tool for live investigations.
- **Directory Export:** Export file lists or hash lists from an active drive or image.

5.1.2 Step-by-Step Forensic Imaging Workflow with FTK Imager

1. **Connect Write Blocker:** Connect the target suspect drive to the forensic workstation through a hardware write blocker.
2. **Launch FTK Imager:** Click **File** $\rightarrow$ **Create Disk Image**.
3. **Select Source Type:** Choose whether the source is a *Physical Drive* (entire drive including unallocated space), *Logical Drive* (a single partition), *Image File*, or a *Contents of a Folder*.
4. **Select Source Device:** Choose the correct disk identifier (double-check the size and model to prevent selecting the forensic workstation's boot drive).
5. **Configure Image Destination:**
   - Click **Add** to select the destination.
   - Select format (typically `E01` for cases requiring comprehensive metadata and integrity tracking).
   - Fill out **Evidence Item Information** (Case Number, Evidence Number, Unique Description, Examiner Name, and custom Notes).
   - Choose the **Destination Folder** and assign an **Image Filename**.
   - Configure fragmentation/segmentation limits (e.g., split image files every 1500MB to fit on smaller fat32 systems, or set to 0 to keep it as one file).
   - Ensure **"Verify images after they are created"** is checked. This forces FTK Imager to recalculate hashes of both the source and the written image to confirm mathematical parity.
6. **Execute & Log:** Click **Start**. Monitor the progress. Once completed, a Verification Summary window displays MD5 and SHA-1 hash comparison values and notes any bad sectors.

5.2 dd (Data Duplicator)

The `dd` command is a standard Unix/Linux command-line utility used for copying and converting files. In forensic acquisitions, it is often utilized to make bit-stream raw images of hard drives or partitions.

5.2.1 Command Syntax & Arguments

Because `dd` is a powerful utility that overwrites disks with zero confirmation prompts, it is jokingly referred to as "disk destroyer". Great caution must be exercised when typing command flags.

The standard syntax for a forensic bit-stream acquisition using `dd` is:

```
dd if=/dev/sda of=/evidence/suspect_disk.raw bs=64K conv=noerror,sync status=progress
```

5.2.2 Parameter Breakdown

- `if=/dev/sda` (*Input File*): Specifies the source device. In Linux systems, raw storage drives are represented as device files (e.g., `/dev/sda`, `/dev/sdb`).
- `of=/evidence/suspect_disk.raw` (*Output File*): Specifies the path where the bit-stream raw image will be saved.
- `bs=64K` (*Block Size*): Configures the size of data chunks read and written at one time. 64KB blocks increase efficiency.
- `conv=noerror,sync` (*Conversion Options*):
  - `noerror`: Instructs the system to continue copying even if it encounters unreadable/bad sectors (otherwise, the tool stops immediately, failing the acquisition).
  - `sync`: Pads missing or corrupted blocks with null bytes (`0x00`) to maintain perfect spatial alignment of subsequent data, ensuring physical offsets remain correct.
- `status=progress`: Displays a live transfer rate and progress bar while copying.

5.2.3 dcfldd: The Forensic Alternative

The standard `dd` tool has a limitation: it does not calculate cryptographic hashes on-the-fly, requiring the examiner to calculate the source hash beforehand and target hash afterward manually. To address this, the Defense Computer Forensics Laboratory (DCFL) created `dcfldd`, an enhanced version of `dd` optimized for digital forensics. Key features of `dcfldd` include:

- **On-The-Fly Hashing:** Computes MD5, SHA-1, or SHA-256 hashes as data is read.
- **Status Progress:** Provides a native, clear progress bar.
- **Multi-output Writing:** Allows writing to multiple identical destination drives simultaneously.

Example command for `dcfldd`:

```
dcfldd if=/dev/sda of=/evidence/suspect.raw hash=sha256 hashlog=/evidence/hash.log
```

6 Student Review & Scenario Questions

Review Questions for Practice

1. **Question 1 (Concept Verification):** Explain why a forensic examiner should avoid booting a suspect's computer directly to look for evidence files. What alternative procedure should be followed?
2. **Question 2 (Chain of Custody):** Imagine you are an investigator. You recovered a USB flash drive from an office, imaged it on-site, and then shipped the physical drive to the headquarters lab. Outline the Chain of Custody entries that must occur during this timeline.
3. **Question 3 (Acquisition Choice):** You are dealing with an employee suspected of corporate espionage. The employee's desktop computer is currently on, and you notice a background process running. How would your acquisition approach differ if the drive was fully encrypted with BitLocker versus unencrypted?
4. **Question 4 (Imaging vs. Cloning):** A client requests you to build a bootable duplicate drive of a compromised server to swap in and minimize server down-time immediately. Should you perform "disk imaging" or "disk cloning"? Explain your technical reasoning.
5. **Question 5 (dd Command Mechanics):** Under what conditions would a digital forensics analyst append the options `conv=noerror,sync` to a standard Unix `dd` command? What consequence occurs if these options are neglected on a damaged hard drive?

Quick Reference Cheat-sheet for Students

- **Rule of Thumb 1:** Always document the scene *before* touching any hardware.
- **Rule of Thumb 2:** Preserve the volatile memory first (Order of Volatility!).
- **Rule of Thumb 3:** Never conduct forensic analysis directly on original evidence; always make a bit-stream copy and work on the duplicate.
- **Rule of Thumb 4:** Calculate and verify cryptographic hash values (MD5, SHA-256) of both the original media and the forensic copy to ensure 100% mathematical parity.